Irfanul Hoda

Mohali, Punjab, India
Github | **LinkedIn** | **LeetCode** | **GeeksForGeeks** | **+91-7209425943** | **irfanulhoda2@gmail.com**

Summary

CSE graduate focused on backend development with Java, Spring Boot, Hibernate, REST APIs, and Docker. Built real-time applications using WebSocket/STOMP and recently developed a Retrieval-Augmented Generation (RAG) chatbot using Generative AI concepts. Strong foundation in DSA, databases, backend development, and Generative AI, with hands-on experience building practical software projects. Seeking an opportunity to contribute to real-world software projects and grow as a backend developer.

Technical Skills and Tools

- **Languages:** C++, Java, Python, SQL
- **Backend / Frameworks:** Spring Boot, Maven, Spring Security, Hibernate, REST APIs
- **Generative AI:** RAG, OpenAI API, LLMs, Prompt Engineering, Embeddings, Vector Retrieval
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Tools:** Git, GitHub, Postman, VS Code, Docker, IntelliJ IDEA, Jupyter Notebook
- **CS Fundamentals:** OOP, Computer Networks, Operating Systems, System Design, DBMS, DSA

Projects

Real-Time Chat Application | WebSocket · Messaging Java · Spring Boot · WebSocket (STOMP) · PostgreSQL · REST APIs

- Engineered a real-time messaging system using Spring Boot WebSocket with STOMP, supporting multi-room broadcast and private messaging with sub-second message delivery latency.
- Persisted chat history in PostgreSQL with a paginated REST endpoint for message retrieval, enabling conversation continuity across sessions with ∼60% faster load vs. full-history fetch.

Persona-Matched AI Chatbot (RAG + LLM) Java · PostgreSQL (pgvector) · Docker · Ollama · Claude API · Telegram Bot API

- Built an end-to-end RAG pipeline that replicates personal conversational style, parsing 5,000+ real chat messages into embeddings via Ollama and storing them in PostgreSQL with pgvector for semantic similarity search.
- Designed a retrieval-augmented generation system combining few-shot example retrieval (cosine similarity) with a static fact-grounding layer, reducing LLM hallucination by filtering low-relevance matches below a tuned similarity threshold.
- Containerized the vector database and embedding model with Docker Compose, and integrated Claude (Anthropic API) to generate persona-matched replies; deployed as a live Telegram bot handling real-time conversational traffic.

Technical Practice

Full-Stack Web Development (Spring Boot, REST APIs, System Design) January 2025 – Present

- Developed backend applications using Spring Boot and REST APIs.

Data Structures & Algorithms (DSA) with Java January 2025 – Present

- Solved 100+ problems on LeetCode covering arrays, strings, recursion, trees, graphs, and DP.
- Strengthened algorithmic thinking and time-space complexity analysis through structured practice.

Certifications

- **Machine Learning - NPTEL, IIT(KGP)**
- **Deep Learning - NPTEL, IIT(KGP)**
- **Fundamental of Java Programming - Coursera**
- **Python for Everybody**
- **React Native**
- **Wire Less Communications of Everybody**
- **Networking in Google Cloud: Network Architecture**
- **Introduction to the Internet of Things and Embedded Systems**

Achievements

- **Secured 3rd position in Pharmathon among 80+ teams.**

Education

Chandigarh University, Mohali, Punjab June 2022 – May 2026
B.E in Computer Science and Engineering – CGPA: 7/10

Good Samaritan School, Jasola, Sarita Vihar, New Delhi Apr 2018 – Mar 2020
Senior Secondary (Class XII) – 82.8%

Holy Mission School, Laheriasarai, Darbhanga, Bihar Apr 2017 – Mar 2018
Matriculation (Class X) – 84.4%

Hobbies & Interests

Coding, Problem-solving, Cricket, Kabaddi, Chess